Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/08/2006
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	A Quantitative Test of On-Orbit Exercise Countermeasures for Bone Demineralization Using a Bed Rest Analog		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cavanagh@u.washington.edu	**Fax:**	FY 206-685-3139
PI Organization Type:	UNIVERSITY	**Phone:**	206-221-2845
Organization Name:	University of Washington		
PI Address 1:	Department of Orthopaedics and Sports Medicine		
PI Address 2:	School of Medicine		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	06/01/2005	**End Date:**	09/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: PI moved to University of Washington in summer 2008 and end date changed to 9/30/2008 (from 5/31/2009); new grant will be issued for remainder of this award, per J. Dardano/JSC (8/2008)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davis, Brian L (Cleveland Clinic) Licata, Angelo A (Cleveland Clinic) Ilaslan, Hakan (Cleveland Clinic)		
Grant/Contract No.:	NNJ05HE72G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bedrest is an excellent analog for the study of on-orbit musculoskeletal changes that occur during long-duration spaceflight. Our novel approach involves quantifying typical daily loads to the lower extremities in selected free-living volunteer subjects who are moderate exercisers and planning individual exercise prescriptions based on daily load stimulus theory. Subjects will then be randomized to control or exercise groups and confined to 6-degree head-down bedrest in the General Clinical Research Center (GCRC) at the Cleveland Clinic Foundation (CCF) for 12 weeks. During this time, the exercise group will undergo individualized daily exercise programs in the unique Zero Gravity Locomotion Simulator (ZLS) at CCF designed to replace their daily mechanical load stimulus experienced during free living. A treadmill and a resistance-exercise device similar to those available on the International Space Station will be used for exercise. DXA, MRI, and QCT scans will be conducted at the start and end of bedrest to assess variations in bone mineral density, changes in muscle volume, and regional changes (cortical vs. trabecular) in bone. Strength measurements will be collected pre- and post-bedrest, and urinary and serum markers of bone resorption and formation will also be assessed. Dietary intake sufficient to maintain constant body weight with a balanced intake of macro- and micronutrients will be planned and supervised by a registered dietician, and a physician will be available 24 hr/day to attend to any medical needs. A rehabilitation program will be offered to subjects in the 8 weeks following bedrest. The proposed experiment should provide a categorical answer to the question of whether intermittent load replacement can adequately protect the musculoskeletal system against hypokinetic osteopenia and muscle atrophy and may open a new era in individualized exercise countermeasure planning. The specific aim of this project is to determine if individualized mechanical load replacement and supplementation during bedrest can serve as a countermeasure to losses in BMD, bone quality, and muscle atrophy during 12 weeks of bedrest. The goals of the proposed study are congruent with the goals identified in the critical path roadmap for countermeasure development. We will address the following Enabling Question 1e of the Bioastronautics Roadmap: What are the specifics of the optimal exercise regimen with regard to mode, duration, intensity and frequency, to be followed during exposure to hypogravity so as to minimize decreases in bone mass? Is impact loading an essential element and, if so, how can it be produced in hypogravity?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information obtained as a result of this study will provide further insight into the role of skeletal loading in bone loss during long-duration spaceflight, and knowledge gained from this research will thereby contribute to a better understanding of the importance of exercise for the development and maintenance of bone strength among humans living in Earth gravitational fields. Knowledge in this area is crucial to the treatment of bone disease for which exercise may or may not be an effective intervention.
Task Progress:	The previous fiscal year has been very productive for this project. During this time GCRC support was obtained and an infrastructure was established to carry out the research. The first campaign, which included two male control subjects, has been successfully completed. The second campaign is currently underway and involves two female subjects, one control and one exercising subject. In collaboration with an NSBRI-funded project, the NASA Glenn Research Center and CCF project team developed a passive pneumatic subject load device (pSLD). This device uses a single pneumatic cylinder to provide a nearly constant spring force on the cables attached to the subject harness. Proof of concept testing was successfully completed at GRC and a duplicate system was installed and validated at CCF. This system is currently being used during exercise in Campaign 2. The project team continues the development of the linear motor subject load device and a bungee system similar to that used on the International Space Station. The project team has modified existing mathematical theories on daily load stimulus (DLS) to develop a comprehensive algorithm. We formulated a revised DLS algorithm which accounts for the peak load as well as the rate of change of the bone load and other important factors. This revised DLS is being used to formulate the individualized exercise prescriptions used by the exercising subject. The revised algorithm is currently being implemented into a toolkit for the real-time processing of test subject force-plate data.
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Papers from Meeting Proceedings	Cavanagh PR, Rice AJ, Davis BL, Carter DR. "A Quantitative Test of On-Orbit Exercise Countermeasures to Bone Demineralization Using a Bedrest Analog." Bioastronautics Investigators' Workshop, Galveston, TX, January 10-12, 2005. POSTER Bioastronautics Investigators' Workshop, January 2005. , Jan-2005